Subtype Entailment over $\{\bot, \times\}$: a self-contained characterization, a corrected reachability lemma, and the border-avoidance problem in occurrence form

Memo for Fritz Henglein

revision of `sep-status-and-attack.tex` for the $\top$-free signature,
incorporating the August 27 analysis of border avoidance
building on `characterization.tex`, `pda.tex`, `pda3.tex`, `ent.tex` (Henglein–Rehof–Urzyczyn, 2003)

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Abstract

This is a revision of the August 14 memo for the signature $\Sigma = \{\bot^{(0)}, \times^{(2)}\}$ — $\top$ removed as a constructor — with types the possibly infinite Σ-trees ordered by $t_1 \leq t_2 \iff \mathrm{Paths}(t_1) \subseteq \mathrm{Paths}(t_2)$. Two things change and one does not.

What changes, and simplifies. A type is nothing but the prefix-closed set of its $\times$-nodes, and $\leq$ is inclusion; the type poset is a complete distributive lattice whose greatest element is the *type* $\Omega = \mu t.\, t \times t$, so deleting the constructor $\top$ does not delete a top element from the order — it deletes a second *label* that has to be propagated. Consequently the whole two-machine apparatus of `pda3.tex` (dual machines, safety, the permutation lemma, clash exclusion, conditions (a)–(h)) collapses to a *single* $\times$-propagation pushdown system $\mathcal{W}$, and the path characterization becomes a four-line Horn-logic argument (Theorem 2.6): all constraints are closure rules, the canonical countermodel is the least solution above a seed, and the three side conditions read off directly. This completes, and corrects by one condition, the programme begun in `characterization.tex` (November 2003), whose soundness theorem is one half of Lemma 2.3 and whose "canonical substitution" idea is Theorem 2.6. All of it is machine-checked against an exact model enumeration (300 random systems; 0 failures on four separate checks).

What does not change. The residual object is the same. The single surviving condition is exactly the prefix-reach condition (b) of `pda3.tex` on the machine $\mathcal{A}_\bot = \mathcal{W}^{\mathrm{rev}}$, whose regularity (Lemma 3, *taminazad*) is false — and the counterexample is realized, in the $\top$-free system, by the same one-constraint system $\{\alpha \times \beta \leq \alpha\}$, now with the query $\beta \leq \alpha$ (Theorem 3.2; machine-verified). So SEP over $\{\bot, \times\}$ reduces to one instance of prefix-reach avoidance, hence to border avoidance (BorderAvoid): can a word avoid automaton-flagged borders in all of its prefixes?

New: BorderAvoid *in occurrence form (§8).* Indexing the agreement condition by *two* coordinates — the length b of the repeated prefix and the period d it induces — rather than one gives, for each pair (L, D), two finite-state conditions $N''(L, D)$ (necessary) and $S''(L, D)$ (sufficient) with $S'' \Rightarrow$ witness $\Rightarrow N''$, and every witness satisfies $S''(\mathrm{lrs}(\pi), 0)$. Hence BorderAvoid is decidable *if and only if* the least budget $L + D$ certifying an existing witness is bounded by a computable function of the instance (Theorem 8.9): the open problem becomes a bound on one integer, and the decision procedure becomes a finite-automaton emptiness test. A creation-free *truncation theorem* (Theorem 8.15)

reduces that bound further, to a statement about two runs of one DFA never merging. The problem itself remains open.

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1 The problem

1.1 Types are prefix-closed sets of paths

Fix $\Sigma = \{\bot^{(0)}, \times^{(2)}\}$ with $\times$ binary and covariant.

Definition 1.1 (types)**.** A *type* is a partial map $t : \{0,1\}^* \rightharpoonup \Sigma$ whose domain $\mathrm{Paths}(t)$ is nonempty and prefix-closed and which respects arities: $t(p) = \times$ implies $p0, p1 \in \mathrm{Paths}(t)$, and $t(p) = \bot$ implies $p0, p1 \notin \mathrm{Paths}(t)$. Types may be infinite. Write

$$\mathrm{X}(t) \;:=\; \{\, p : t(p) = \times \,\}$$

for the set of *internal* ($\times$-labelled) nodes.

Definition 1.2 (order)**.** $t_1 \le t_2$ iff for every p occurring in t_1 with $t_1(p) = \times$, p occurs in t_2 and $t_2(p) = \times$; that is, $\mathrm{X}(t_1) \subseteq \mathrm{X}(t_2)$.

Lemma 1.3 (the three readings of $\le$ coincide)**.** *For all types t_1, t_2 the following are equivalent:*

(i) $\mathrm{X}(t_1) \subseteq \mathrm{X}(t_2)$*;*

(ii) $\mathrm{Paths}(t_1) \subseteq \mathrm{Paths}(t_2)$*;*

(iii) $t_1 \le t_2$ *in the coinductive order generated by* $\bot \le t$ *and* $t_1 \times t_2 \le t_1' \times t_2'$ *whenever* $t_i \le t_i'$*.*

Moreover $t \mapsto \mathrm{X}(t)$ is an order-isomorphism from types onto the prefix-closed subsets of $\{0,1\}^*$ ordered by inclusion, with $\mathrm{X}(\bot) = \emptyset$ and $\mathrm{X}(t_1 \times t_2) = \{\varepsilon\} \cup 0\mathrm{X}(t_1) \cup 1\mathrm{X}(t_2)$.

Proof. $\mathrm{Paths}(t) = \{\varepsilon\} \cup \mathrm{X}(t) \cdot \{0,1\}$, so (i)⇒(ii). For (ii)⇒(i): if $t_1(p) = \times$ then $p0 \in \mathrm{Paths}(t_1) \subseteq \mathrm{Paths}(t_2)$, which forces $p \in \mathrm{Paths}(t_2)$ and $t_2(p) \neq \bot$ (a $\bot$ has no children), i.e. $t_2(p) = \times$. (i)⇔(iii) by induction on the depth of p in one direction and coinduction in the other: $\mathrm{X}(t)$ is prefix-closed, and $t_1 \le_{\mathrm{coind}} t_2$ holds iff no $\times$ of t_1 meets a $\bot$ or an absent node of t_2. $\mathrm{X}(t)$ is prefix-closed because a node above a $\times$ is internal; conversely a prefix-closed X defines the type with $\times$ on X and $\bot$ on $X\{0,1\} \setminus X$. The displayed identities are immediate. □

Corollary 1.4 (the lattice, and where $\top$ went)**.** *The type order is a complete distributive lattice: meets and joins are intersections and unions of* X*-sets. Its least element is $\bot$ and its greatest element is the* type $\Omega = \mu t.\, t \times t$*, with $\mathrm{X}(\Omega) = \{0,1\}^*$. Thus removing $\top$ from Σ removes neither the top of the order nor a lattice structure; what it removes is a second constant* label *that solutions must propagate.*

Remark 1.5. This is why the development below is one-sided. With $\top$ present, a solution has to place two competing labels and the 2003 programme needs two machines $\mathcal{A}_\bot, \mathcal{A}_\top$, a safety analysis, and a clash-exclusion ("permutation") lemma to keep them apart. Here a solution places one label, "$\times$", and everything not forced to be $\times$ can be $\bot$.

1.2 Constraints and the entailment problem

Definition 1.6. A *constraint set* C is a finite set of inequalities between Σ-terms over variables. Introducing fresh variables for subterms (with each fresh variable equated to its subterm by two inequalities, a conservative extension) every C normalizes to constraints of the four forms

$$\alpha \times \beta \le \gamma, \qquad \alpha \le \beta \times \gamma, \qquad \alpha \le \beta, \qquad \alpha \le \bot,$$

$\bot \le \alpha$ being vacuous. A substitution S maps variables to types; $S \models C$ means S satisfies every constraint, and $C \models \alpha \le \beta$ means every $S \models C$ satisfies $S(\alpha) \le S(\beta)$.

Problem 1.7 (SEP over $\{\bot, \times\}$)**.** Given C, α, β, decide $C \models \alpha \le \beta$.

Lemma 1.8 (constraints are inclusions)**.** *Write* $X_v := \mathrm{X}(S(v))$. *Then* $S \models C$ *iff for every constraint of* C*:*

$$\begin{array}{ll} \alpha \times \beta \le \gamma: & \varepsilon \in X_\gamma, \quad 0X_\alpha \subseteq X_\gamma, \quad 1X_\beta \subseteq X_\gamma; \\ \alpha \le \beta \times \gamma: & X_\alpha/0 \subseteq X_\beta, \quad X_\alpha/1 \subseteq X_\gamma \quad (\textit{where } X/a = \{q : aq \in X\}); \\ \alpha \le \beta: & X_\alpha \subseteq X_\beta; \\ \alpha \le \bot: & X_\alpha = \emptyset. \end{array}$$

Proof. Immediate from Lemma 1.3 and $\mathrm{X}(\beta \times \gamma) = \{\varepsilon\} \cup 0X_\beta \cup 1X_\gamma$; for the second line, $X_\alpha \subseteq \{\varepsilon\} \cup 0X_\beta \cup 1X_\gamma$ says exactly that every nonempty $p \in X_\alpha$ starting with a has its tail in the a-component. □

Every clause of Lemma 1.8 is a Horn closure rule "$q \in X_u \Rightarrow f(q) \in X_v$" for f one of $q \mapsto q$, $q \mapsto aq$, $aq \mapsto q$, together with the unconditional facts $\varepsilon \in X_\gamma$ and the denials $X_\alpha = \emptyset$. That observation is the whole of §2.

1.3 Known results

Satisfiability of non-structural subtype constraints is decidable in cubic time (Palsberg–O'Keefe; Pottier); Proposition 2.5 below re-derives it for this fragment in two lines. For entailment: structural/finite is coNP-complete and structural/recursive PSPACE-complete (Henglein–Rehof, LICS'97, ICALP'98); non-structural entailment is PSPACE-hard over both finite and infinite trees (ICALP'98) and its decidability has been open since Trifonov–Smith and Pottier raised it in 1996. The first-order *theory* of non-structural subtyping is undecidable (Su–Aiken–Niehren–Priesnitz–Talbot, POPL'02), but entailment is a Π_1 fragment the undecidability proof does not reach. Niehren–Priesnitz (TACS'01; Inf. & Comp. 2003) characterized entailment for one constructor by universality of *cap set expressions* and settled the unary case; the general problem is, on a fresh 2026 literature check, still open. The $\top$-free fragment treated here is the one begun in `characterization.tex` (Nov. 2003), which proves the soundness half of Lemma 2.3 and sketches — without the side conditions — the canonical substitution of Theorem 2.6.

2 The $\times$-propagation machine and the characterization

2.1 One machine

Definition 2.1 ($\mathcal{W}$)**.** Let C be normalized. The pushdown system $\mathcal{W} = \mathcal{W}(C)$ has the variables of C as states, stack alphabet $\{0, 1\}$ (top of stack = leftmost letter = step nearest the root), and the moves

$$\begin{array}{llll} \alpha \vdash \gamma, 0 & \beta \vdash \gamma, 1 & \text{for } (\alpha \times \beta \le \gamma) \in C & \text{(pushes)} \\ \alpha, 0 \vdash \beta & \alpha, 1 \vdash \gamma & \text{for } (\alpha \le \beta \times \gamma) \in C & \text{(pops)} \\ \alpha \vdash \beta & & \text{for } (\alpha \le \beta) \in C & (\varepsilon) \end{array}$$

Configurations are $\langle v, q\rangle$, $q \in \{0,1\}^*$; $\to^*$ is reachability. Put

$$\mathrm{BF} := \{\gamma : (\alpha \times \beta \le \gamma) \in C \text{ for some } \alpha, \beta\}, \qquad I_\bot := \{\delta : (\delta \le \bot) \in C\}.$$

Remark 2.2. $\mathcal{W}$ is exactly $\mathcal{A}_\bot$ of `pda3.tex` with every move reversed (equivalently, $\mathcal{A}_\top$, now read as $\times$-propagation rather than $\top$-propagation). There is only one machine because there is only one label to propagate.

Lemma 2.3 (soundness of propagation)**.** *If $\langle u, q\rangle \to^* \langle v, q'\rangle$ then in every $S \models C$: $q \in X_u$ implies $q' \in X_v$.*

Proof. One move at a time, by Lemma 1.8: a push move $u \vdash v, a$ comes from a constraint whose clause is $aX_u \subseteq X_v$; a pop move $u, a \vdash v$ from $X_u/a \subseteq X_v$; an ε-move from $X_u \subseteq X_v$. □

Lemma 2.3 is the contrapositive of the "bottom-propagation" lemma of `characterization.tex`: there, $S \models \gamma, \pi$ means some prefix of π carries $\bot$ in $S(\gamma)$, which is precisely $\pi \notin X_\gamma$; the transition system of that note is $\mathcal{W}$ with all moves reversed.

2.2 Least solutions

Lemma 2.4 (closure lemma)**.** *Let $Y = (Y_v)$ be a family of prefix-closed sets of paths and let*

$$\mathrm{Cl}(Y) \;:=\; \big(\textit{ configurations } \to^*\textit{-reachable from } \{\langle v, q\rangle : q \in Y_v\} \cup \{\langle \gamma, \varepsilon\rangle : \gamma \in \mathrm{BF}\}\big),$$

read as a family $\mathrm{Cl}(Y)_v = \{q : \langle v, q\rangle \in \mathrm{Cl}(Y)\}$. Then every $\mathrm{Cl}(Y)_v$ is prefix-closed, $\mathrm{Cl}(Y)$ satisfies every constraint of C except possibly the denials, and if $\mathrm{Cl}(Y)_\delta = \emptyset$ for every $\delta \in I_\bot$ then $\mathrm{Cl}(Y)$ is the least solution of C above Y.

Proof. Write $\mathrm{Cl}(Y) = \bigcup_n Z^n$ where Z^0 is Y together with the facts $\varepsilon \in Z^0_\gamma$ ($\gamma \in \mathrm{BF}$), and Z^{n+1} adds one round of rule applications. Z^0 is prefix-closed. If Z^n is, so is Z^{n+1}: the ε-rule adds Z^n_u to Z^{n+1}_v; the push rule adds aZ^n_u to Z^{n+1}_v and $\varepsilon \in Z^0_v$ already, and $\{\varepsilon\} \cup aP$ is prefix-closed when P is; the pop rule adds Z^n_u/a, and P/a is prefix-closed when P is. A union of prefix-closed sets is prefix-closed. Closure under the rules is exactly satisfaction of the non-denial clauses of Lemma 1.8, the facts $\varepsilon \in X_\gamma$ being supplied by Z^0. Minimality: any solution above Y contains Z^0 (Lemma 1.8) and is closed under the rules. □

Proposition 2.5 (satisfiability)**.** *C is satisfiable iff no $\delta \in I_\bot$ is reachable in $\mathcal{W}$ from any $\langle \gamma, \varepsilon\rangle$ with $\gamma \in \mathrm{BF}$ (in particular $I_\bot \cap \mathrm{BF} = \emptyset$). This is decidable in polynomial time, and the least solution is $L := \mathrm{Cl}(\emptyset)$.*

Proof. If some such δ is reachable then $X_\delta \neq \emptyset$ in every solution by Lemma 2.3 (using $\varepsilon \in X_\gamma$, forced by Lemma 1.8), contradicting $\delta \leq \bot$. Otherwise $L = \mathrm{Cl}(\emptyset)$ is a solution by Lemma 2.4. Reachability of a state from a configuration with empty stack is the emptiness of a regular push-summary language. □

2.3 The characterization

Theorem 2.6 (path characterization, $\top$-free)**.** *Let C be normalized and satisfiable. Then $C \not\models \alpha \leq \beta$ if and only if there is a path $p \in \{0,1\}^*$ such that*

(S) for every $\delta \in I_\bot$ and every $p' \sqsubseteq p$: $\langle \alpha, p'\rangle \not\to^* \langle \delta, q\rangle$ for all q;

(F) $\langle \gamma, \varepsilon\rangle \not\to^* \langle \beta, p\rangle$ for every $\gamma \in \mathrm{BF}$;

(B) $\langle \alpha, p'\rangle \not\to^* \langle \beta, p\rangle$ for every $p' \sqsubseteq p$.

In that case the family

$$X_v \;=\; \{\, q : \langle \alpha, p' \rangle \to^* \langle v, q \rangle \text{ for some } p' \sqsubseteq p \,\} \;\cup\; \{\, q : \langle \gamma, \varepsilon \rangle \to^* \langle v, q \rangle \text{ for some } \gamma \in \mathrm{BF} \,\}$$

is a countermodel, and it assigns regular *trees.*

Proof. ($\Leftarrow$) Let $Y_\alpha := \{p' : p' \sqsubseteq p\}$ and $Y_v := \emptyset$ otherwise; Y is prefix-closed and $\mathrm{Cl}(Y)$ is the displayed family. By (S) it meets no denial, so by Lemma 2.4 it is a solution. It contains p in X_α, and by (B) and (F) it does not contain p in X_β. Hence $C \not\models \alpha \le \beta$.

($\Rightarrow$) Let $S \models C$ with $p \in X_\alpha \setminus X_\beta$ for some p. (S): if $\langle \alpha, p' \rangle \to^* \langle \delta, q \rangle$ with $p' \sqsubseteq p$ and $\delta \in I_\perp$ then $p' \in X_\alpha$ (X_α is prefix-closed) and Lemma 2.3 gives $q \in X_\delta = \emptyset$, absurd. (B): if $\langle \alpha, p' \rangle \to^* \langle \beta, p \rangle$ with $p' \sqsubseteq p$ then Lemma 2.3 gives $p \in X_\beta$, contrary to choice. (F): if $\langle \gamma, \varepsilon \rangle \to^* \langle \beta, p \rangle$ with $\gamma \in \mathrm{BF}$ then $\varepsilon \in X_\gamma$ (Lemma 1.8) and Lemma 2.3 gives $p \in X_\beta$, again contrary to choice.

Regularity: the set of configurations reachable from a regular set of configurations of a pushdown system is regular and effectively computable, and $\{\langle \alpha, p' \rangle : p' \sqsubseteq p\} \cup \{\langle \gamma, \varepsilon \rangle\}$ is regular; so each X_v is a regular language, i.e. $S(v)$ is a regular tree. □

Corollary 2.7. *$C \models \alpha \le \beta$ over arbitrary trees iff over regular trees. Countermodels can always be taken regular, with a description of size polynomial in $|C|$ and $|p|$.*

Corollary 2.8 (the two easy conditions are regular)**.** *For fixed C, α, β each of* (S) *and* (F) *defines a regular set of paths p, effectively computable.*

Proof. (F): $\{p : \langle \gamma, \varepsilon \rangle \to^* \langle \beta, p \rangle\} = L_2^{\gamma\beta}$ is regular (push summaries, `pda.tex` Lemma 2), and (F) is the complement of a finite union. (S): $U := \{p' : \langle \alpha, p' \rangle \to^* \langle \delta, q \rangle$ for some $\delta \in I_\perp, q\}$ is pre* of a regular set of configurations intersected with the α-slice, hence regular; (S) says $p \notin U\{0,1\}^*$, again regular. □

Remark 2.9 (what is new relative to 2003)**.** `characterization.tex` proves Lemma 2.3 (in its dual, $\perp$-propagating form) and states the ($\Leftarrow$) direction under the hypothesis (B) alone, with the canonical substitution left as an "idea". Two conditions are missing there and are genuinely needed: (S), without which the canonical substitution can violate a $\delta \le \perp$ constraint, and (F), without which the path p may already be forced into X_β by the unconditional facts $\varepsilon \in X_\gamma$ ($\gamma \in \mathrm{BF}$). Example for (F): $C = \{\alpha \times \gamma \le \beta\}$ and $p = \varepsilon$ satisfies (S) and (B) but $\varepsilon \in X_\beta$ always, and indeed $C \models \alpha \le \beta$ fails only at $p = 1$, not at $p = \varepsilon$.

Theorem 2.10 (reduction to prefix-reach avoidance)**.** *Define*

> PrefixAvoid*: given a pushdown system $\mathcal{A}$ (push/pop/ε), states β, α and a regular $R \subseteq \{0,1\}^*$, decide whether some $\pi \in R$ satisfies $\langle \beta, \pi \rangle \not\to^* \langle \alpha, \pi' \rangle$ for every $\pi' \sqsubseteq \pi$.*

Then SEP *over $\{\perp, \times\}$ is decidable if* PrefixAvoid *is. Conversely the transition structure of* PrefixAvoid *is fully realized: every pushdown system arises as $\mathcal{W}(C)$ for a constraint set C, up to harmless padding.*

Proof. Forward: by Theorem 2.6 and Corollary 2.8, $C \not\models \alpha \le \beta$ iff some p in the regular set (S)$\cap$(F) satisfies (B); and (B) is the PrefixAvoid condition for the reversed machine, since $\langle \alpha, p' \rangle \to^*_{\mathcal{W}} \langle \beta, p \rangle$ iff $\langle \beta, p \rangle \to^*_{\mathcal{W}^{\mathrm{rev}}} \langle \alpha, p' \rangle$, and $\mathcal{W}^{\mathrm{rev}} = \mathcal{A}_\perp$. Converse (transition structure): a lone pop $\rho, 0 \vdash \gamma$ of $\mathcal{A}_\perp$ is the $\mathcal{W}$-push $\gamma \vdash \rho, 0$, realized by $\gamma \times \varphi \le \rho$ with φ fresh; a

lone push $\gamma \vdash \rho, 0$ of $\mathcal{A}_\perp$ is the $\mathcal{W}$-pop $\rho, 0 \vdash \gamma$, realized by $\rho \le \gamma \times \psi$ with ψ fresh; ε-moves directly. Matching an arbitrary regular R exactly against the specific shape of (S)∩(F) is the remaining (routine but unfinished here) step of a full converse. □

Remark 2.11 (the side conditions are not decoration)**.** It is tempting to hope that (S) and (F) can be absorbed into the machine and PrefixAvoid replaced by its plain, unannotated version. They cannot: for a pushdown system with push/pop/ε moves and states $a \neq b$, the plain question "is there an s with no $s' \sqsubseteq s$ such that $\langle a, s\rangle \to^* \langle b, s'\rangle$", with no regular restriction on s, holds iff $\langle a, \varepsilon\rangle \not\to^* \langle b, \varepsilon\rangle$ — polynomial time (P27 of `STATUS.md`, 30 August 2026: a run $\langle a, \varepsilon\rangle \to^* \langle b, \varepsilon\rangle$ pops only its own pushes, hence lifts over every s; otherwise $s = \varepsilon$ already works). All the difficulty of PrefixAvoid, and so of sep, sits in the interaction between the machine and the regular annotation.

2.4 Machine-checked validation

Script `topfree.py` implements $\mathcal{W}$, the three conditions and the canonical model; `verify_topfree.py` checks Theorem 2.6 against an exact model enumeration. Ground truth: a type is presented by the antichain B of its $\perp$-leaves, so $\mathrm{X} = \{q :$ no prefix of q lies in $B\}$; this family contains $\perp$ ($B = \{\varepsilon\}$), Ω ($B = \emptyset$), every finite tree, and infinitely many infinite ones, and all constraint checks on it are exact set operations. Over 300 random normalized systems (2–3 variables, 2–4 constraints, B ranging over all 26 antichains of depth ≤ 2, witness paths to length 5, canonical models checked to depth 5):

check	cases	failures
(C1) brute-force countermodel ⇒ Theorem 2.6 predicts non-entailment	144	0
(C2) predicted non-entailment ⇒ canonical model is prefix-closed, satisfies C, separates	150	0
(C3) predicted satisfiable ⇒ the least solution really satisfies C	264	0
(C4) brute-force model exists ⇒ Proposition 2.5 says satisfiable	263	0

(C2) and (C3) are the strong checks: they verify the constructed model itself and need no enumeration family, so they scale. Re-run on larger systems (2–5 variables, 2–7 constraints, witness paths to length 6, models checked to depth 6): 456 canonical countermodels and 734 least models verified, 0 failures. Theorem 3.2 is checked the same way: for $C_0 = \{\alpha \times \beta \le \alpha\}$ and the query $\beta \le \alpha$ the procedure returns the witness $p = 10$ and the countermodel $\mathrm{X}_\beta = \{\varepsilon, 1, 10\}$, $\mathrm{X}_\alpha = \{\varepsilon, 0, 1, 00, 01, 11, \dots\} \not\ni 10$.

3 Refutation of the regularity lemma

Condition (B) of Theorem 2.6 is, by the last proof, membership in

$$L^{\beta\alpha} \;=\; \{\, \pi : \langle \beta, \pi\rangle \to^*_{\mathcal{A}} \langle \alpha, \pi'\rangle \text{ for some } \pi' \sqsubseteq \pi \,\}, \qquad \mathcal{A} := \mathcal{W}^{\mathrm{rev}} = \mathcal{A}_\perp.$$

The 2003–04 programme intended to decide it by the following lemma.

Lemma 3.1 (Lemma 3 of `pda.tex`/`pda3.tex`, **false** as stated)**.** *For all states α, β of an npda, $L^{\alpha\beta}$ is regular, and equals $\big(\bigcup_\gamma L_1^{\alpha\gamma} \cap (L_2^{\gamma\beta} \cdot \Sigma^*)\big)\Sigma^*$ where $L_1^{\alpha\gamma} = \{w : \langle \alpha, w\rangle \to^* \langle \gamma, \varepsilon\rangle\}$ and $L_2^{\gamma\beta} = \{w : \langle \gamma, \varepsilon\rangle \to^* \langle \beta, w\rangle\}$.*

Theorem 3.2 (counterexample, realized $\top$-freely)**.** *Let $C_0 = \{\alpha \times \beta \le \alpha\}$, a satisfiable one-constraint system over $\{\bot, \times\}$. Then $\mathcal{W}(C_0)$ has exactly the two pushes $\alpha \vdash \alpha, 0$ and $\beta \vdash \alpha, 1$, so $\mathcal{A} = \mathcal{W}^{\mathrm{rev}}$ has exactly the two pops $\alpha, 0 \vdash \alpha$ and $\alpha, 1 \vdash \beta$, and*

$$M \;:=\; \{\, p : \langle \beta, p' \rangle \to^*_{\mathcal{W}} \langle \alpha, p \rangle \text{ for some } p' \sqsubseteq p \,\} \;=\; L^{\alpha\beta}_{\mathcal{A}} \;=\; \{\, 0^j 1 u : j \ge 0,\ u \sqsubseteq (0^j 1)^\omega \,\},$$

so $M \cap 0^*10^*1 = \{0^j 10^j 1 : j \ge 0\}$ and M is not regular. The claimed formula evaluates to $0^*1\Sigma^*$ and contains $0100 \notin M$.

Proof. From $\langle \beta, p' \rangle$ the only move is the push $\beta \vdash \alpha, 1$, reaching $\langle \alpha, 1p' \rangle$; thereafter only $\alpha \vdash \alpha, 0$ applies, giving exactly the configurations $\langle \alpha, 0^j 1 p' \rangle$. So $p \in M$ iff $p = 0^j 1 p'$ for some j and some $p' \sqsubseteq p$, i.e. iff $p = 0^j 1 u$ with $u \sqsubseteq (0^j 1) u$, which by Lemma 4.1 means $u \sqsubseteq (0^j 1)^\omega$. For $p = 0^j 10^k 1$ this forces $k = j$ by the position of the second 1; non-regularity follows by pumping in j. The formula: $L_1^{\alpha\alpha} = 0^*$, $L_1^{\alpha\beta} = 0^*1$, and with no pushes in $\mathcal{A}$, $L_2^{\gamma\delta} = \{\varepsilon\}$ for $\gamma = \delta$ and $\emptyset$ otherwise. Machine-verified: `verify_lemma3.py` (exhaustive to length 11, 4017 disagreements with the formula) and, in the present $\top$-free reading, `topfree.py` (enumeration of M to length 8, giving $M \cap 0^*10^*1 = \{11, 0101, 001001, 00010001\}$). □

Remark 3.3 (where the proof breaks)**.** The inclusion $\subseteq$ of the lemma is the correct low-water-mark argument reused in §4. The inclusion $\supseteq$ is asserted as "obvious" but is not: from $w = w'u$ with $w' \in L_1^{\alpha\gamma}$, $w' = v'z$, $v' \in L_2^{\gamma\beta}$, the constructed run ends in $\langle \beta, v'u \rangle$, and $v'u \sqsubseteq v'zu$ holds *iff* $u \sqsubseteq zu$ — a periodicity side condition absent from the formula. It is exactly the implicit word equation $\pi\nu = \mu\pi$ of Niehren–Priesnitz. The *exact-return* diagonal $L_4^{\alpha\beta} = \{w : \langle \alpha, w \rangle \to^* \langle \beta, w \rangle\}$ *is* regular (Corollary 4.3); the prefix relaxation is precisely what lets periodicity in.

4 The corrected reachability lemma

Throughout, npdas have push/pop/ε moves only; $\mathcal{W}$ and $\mathcal{A}$ are of this form. $L_1^{\alpha\gamma}$ and $L_2^{\gamma\beta}$ are regular (pop-, resp. push-summaries; `pda.tex` Lemmas 1–2, both correct).

Lemma 4.1 (periodicity)**.** *For $u, z \in \Sigma^*$: $u \sqsubseteq zu$ iff $z = \varepsilon$ or $u \sqsubseteq z^\omega$.*

Proof. ($\Leftarrow$) If $u \sqsubseteq z^\omega$ then $zu \sqsubseteq z^\omega$, and of two prefixes of z^ω the shorter is a prefix of the longer. ($\Rightarrow$) If $u \sqsubseteq zu$ and $z \neq \varepsilon$ then by induction $u \sqsubseteq z^k u$ for all k; take $|z^k| \ge |u|$. □

Theorem 4.2 (corrected reach lemma; cap decomposition)**.** *For all states β, α of an npda,*

$$L^{\beta\alpha} \;=\; \bigcup_{\gamma} \{\, vzu \;:\; v \in L_2^{\gamma\alpha},\ vz \in L_1^{\beta\gamma},\ u \sqsubseteq zu \,\}.$$

Proof. ($\subseteq$) Let $\langle \beta, w \rangle \to^* \langle \alpha, v_f \rangle$ with $v_f \sqsubseteq w$ and let $\langle \gamma, u \rangle$ be of minimal stack height in the run. After it the bottom $|u|$ letters are never popped, hence never inspected: erasing them yields $\langle \gamma, \varepsilon \rangle \to^* \langle \alpha, v \rangle$ with $v_f = vu$; symmetrically $w = w'u$ with $\langle \beta, w' \rangle \to^* \langle \gamma, \varepsilon \rangle$. From $v_f \sqsubseteq w$ we get $|v| \le |w'|$ and $v \sqsubseteq w'$; write $w' = vz$. Then $vu \sqsubseteq vzu$ iff $u \sqsubseteq zu$. ($\supseteq$) Lift the L_1-run over the untouched bottom u, then the L_2-run. □

Corollary 4.3. $L_4^{\beta\alpha} = \{w : \langle \beta, w \rangle \to^* \langle \alpha, w \rangle\} = \big(\bigcup_\gamma L_1^{\beta\gamma} \cap L_2^{\gamma\alpha}\big)\Sigma^*$ *is regular.*

Theorem 4.2 is the specialization of a general theorem of Caucal; §6 states the general form, shows that it is what makes the flag alphabet of §5 finite, and proves that it yields nothing beyond that.

So $L^{\beta\alpha}$ is a finite union of a regular part ($z = \varepsilon$) and *cap parts* $\{vzu : (v,z) \in R_\gamma,\ z \neq \varepsilon,\ u \sqsubseteq z^\omega\}$ with $R_\gamma = \{(v,z) : v \in L_2^{\gamma\alpha}, vz \in L_1^{\beta\gamma}\}$ a regular split relation: the Niehren–Priesnitz cap shape, derived inside the Henglein–Rehof–Urzyczyn framework. Nothing in this section uses the signature, so it is unchanged by the removal of $\top$.

5 From prefix-reach to border avoidance

Fix C, α, β and $\mathcal{A} = \mathcal{W}^{\mathrm{rev}}$. Let D_1 be a DFA computing, along a word π, the memberships $\pi[1..i] \in L_1^{\beta\gamma}$ for all γ simultaneously and D_2 likewise for $L_2^{\gamma\alpha}$; write $c(i)$ for the pair of D_1-, D_2-states after $\pi[1..i]$ (the *class* of position i), and define the *flag relation*

$$G(c(i), c(j)) \quad :\Longleftrightarrow \quad \exists \gamma :\ \pi[1..i] \in L_2^{\gamma\alpha} \ \wedge\ \pi[1..j] \in L_1^{\beta\gamma}.$$

Lemma 5.1 (position form)**.** *$\pi \in L^{\beta\alpha}$ iff there are $0 \le i \le j \le n = |\pi|$ with $G(c(i), c(j))$ such that $\pi(i..n]$ has period $j - i$ (period 0 holds always; "period p" means $\pi[k] = \pi[k+p]$ for all $i < k \le n - p$, vacuous when $p \ge n - i$).*

Proof. Instantiate Theorem 4.2 with $v = \pi[1..i]$, $z = \pi(i..j]$, $u = \pi(j..n]$ and Lemma 4.1; the case $i = j$ is the regular part. □

Lemma 5.2 (suffix-agreement form)**.** *With $s_x := \pi(x..n]$: $\pi(i..n]$ has period $j - i$ iff $s_j \sqsubseteq s_i$. Hence*

$$\pi \notin L^{\beta\alpha} \iff \forall\, 0 \le i \le j \le n :\ s_j \sqsubseteq s_i \Rightarrow \neg G(c(i), c(j)).$$

Lemma 5.3 (border form)**.** *Let $\rho := \pi$ reversed and $c'(x) := c(n - x)$. For $a \le b$, the prefix $\rho[1..b]$ has a border of length a iff $s_{n-a} \sqsubseteq s_{n-b}$. Hence $\pi \notin L^{\beta\alpha}$ iff for every prefix $\rho[1..b]$ and every border length a of it (including $a = 0$ and $a = b$), $\neg G(c'(b), c'(a))$.*

Definition 5.4 (BorderAvoid)**.** *Border avoidance*: given a finite class set K, a flag relation $G' \subseteq K \times K$ and a regular (NFA-checkable) set of validly class-annotated words, decide whether there is an annotated word ρ such that no prefix of ρ has a border whose two endpoint classes are G'-flagged.

Theorem 5.5. SEP *over $\{\bot, \times\}$ is decidable if* BorderAvoid *is decidable.*

Proof. Chain Theorem 2.10 with Lemmas 5.1–5.3; the class annotation and the regular set (S)∩(F) of Corollary 2.8 transport to ρ by reversal (regular languages are closed under reversal; annotations are guessed and verified by an NFA). □

Remark 5.6. Lemma 5.3 puts SEP into the heartland of combinatorics on words: borders and periods of prefixes are the objects of the Knuth–Morris–Pratt failure function, and period sets of words are completely characterized (Fine–Wilf; Guibas–Odlyzko). The whole of SEP over $\{\bot, \times\}$ is the question *"can a word avoid flagged borders in all of its prefixes?"*

6 Caucal's theorem: the source of the finite flag alphabet, and its exact limit

Theorem 4.2 is a special case of a general theorem of Caucal, and it is worth saying precisely what the general theorem buys and what it provably does not. The short version: everything Theorem 6.1 implies for this problem is already consumed — by Theorem 4.2 (the cap decomposition) and by Proposition 6.2 (the finiteness of the flag alphabet, which is what makes BorderAvoid a statement about a *finite* class set at all). Past that point it gives nothing, for reasons that are themselves theorems.

6.1 The theorem, in its sharp form

Theorem 6.1 (Caucal [7], Thm. 1.3; proof in [8]). *For any* finite *rewriting system R, the prefix rewriting $\to^*_R$ is a rational transduction, and a transducer is effectively constructible from R.*

The proof establishes the sharper statement quoted in [7, p. 63]: prefix rewriting is *the componentwise concatenation of a recognizable relation with the identity relation*, i.e.

$$\to^* \;=\; \mathcal{K}\cdot\Delta \;=\; \big\{\,(uw,\,vw)\;:\;(u,v)\in\mathcal{K},\;w\in\Sigma^*\,\big\},\qquad \mathcal{K}\in \mathrm{Rec}(\Sigma^*\times\Sigma^*), \tag{1}$$

where Δ is the identity on Σ^* and, by Mezei's theorem, $\mathrm{Rec}(\Sigma^*\times\Sigma^*)$ is exactly the family of *finite* unions of products of regular languages. Such a relation is called *right synchronized*: the two components agree on a common suffix, and differ only on a recognizable head.

For a pushdown system the recognizable head is read off the pop- and push-summaries:

$$\mathcal{K} \;=\; \bigcup_{\gamma\in Q} L_1^{\beta\gamma}\times L_2^{\gamma\alpha}, \tag{2}$$

the common suffix w being the part of the stack never touched, i.e. the stack at the low-water mark. Substituting (2) into the prefix restriction $v_f \sqsubseteq w$ is exactly the proof of Theorem 4.2: that proof *is* Caucal's factorization, carried out by hand for this case.

6.2 (a) What it sharpens: the flag alphabet is finite

Proposition 6.2 (finite flags). *Because $\mathcal{K}$ in (1) is* recognizable*, the index γ in (2) ranges over a finite set. Hence the class $c(i)=(A_i,B_i)$ of a position, with*

$$A_j=\{\gamma:\pi[1..j]\in L_1^{\beta\gamma}\},\qquad B_i=\{\gamma:\pi[1..i]\in L_2^{\gamma\alpha}\},$$

ranges over the finite set $K\subseteq 2^Q\times 2^Q$, each component is computed by a deterministic subset construction along π, and the flag relation of §5 has the "non-empty intersection" shape

$$G\big((A,B),(A',B')\big)\iff A'\cap B\neq\emptyset.$$

This is the one place where the general theorem is genuinely load-bearing, and it is worth stating the counterfactual. Suppose $\to^*$ were merely *rational* rather than of the form $\mathcal{K}\cdot\Delta$ with $\mathcal{K}$ recognizable. Then the decomposition (2) would carry an unbounded index, there would be no finite class set K, no finite flag relation G, and BorderAvoid (Definition 8.1) could not even be *stated* — let alone the entire occurrence-form development of §8, every step of which

$(N''(L,D)$, $S''(L,D)$, the budget $\mathcal{B}$, the good tree) presupposes a finite K. Theorem 6.1 is therefore not an unused tool waiting to be applied to the problem; it is the foundation the reformulation already stands on.

Remark 6.3 (the shape of G is not arbitrary, but that has not yet helped)**.** Proposition 6.2 says more than "K is finite": the flag relations arising from constraint systems form a proper subclass of the arbitrary $G \subseteq K \times K$ allowed in Definition 8.1. It is therefore reasonable to ask whether Conjecture C5 — a computable bound on $\mathcal{B}$ (Theorem 8.9) — might hold for the intersection-shaped subclass even if not in general. Tested (`interestingattempt/caucal_shape.py`): random intersection-shaped instances built from random L_1/L_2 edge relations, $|K|$ up to 53, against arbitrary G of matched density on the same DFA, under adversarial hill-climbing at budget ≤ 3. *No separation was observed*: both families cap at $\mathcal{B} = 1$ over ∼4200 settling calls, with no unsettled instance. Intersection-shaped instances are markedly more often NO (684/1200 versus 343/1200 at $|Q| = 4$) — such flags are harder to satisfy — but the budget is not visibly smaller. The test is not yet decisive, since the generator cannot push the free class above $\mathcal{B} = 1$ either.

6.3 (b) Why it gives nothing beyond: three barriers

The tempting next step is to keep working inside the rational world: intersect $\to^*$ with the prefix relation $\mathrm{Pref} = \{(s, s') : s' \sqsubseteq s\}$ and project. That is what `pushdown_prefix_property_proof.pdf` attempts, and it is exactly the 2004 error of Lemma 3 (§3) in modern dress. Three theorems say it cannot be made to work.

Proposition 6.4 (B1: Pref is rational but not recognizable)**.** Pref *is rational, but* $\mathrm{Pref} \notin \mathrm{Rec}(\Sigma^* \times \Sigma^*)$. *Consequently the Elgot–Mezei closure property "*$\mathrm{Rat} \cap \mathrm{Rec} \subseteq \mathrm{Rat}$*" does not apply to* $\to^* \cap \mathrm{Pref}$.

Proof. Rationality is clear (a transducer copies a prefix and discards the rest). Suppose $\mathrm{Pref} = \bigcup_{i \leq n} R_i \times S_i$. Every pair (w, w) lies in Pref, so some i has infinitely many w with $w \in R_i \cap S_i$. Take two such words $w_1 \neq w_2$. If they are $\sqsubseteq$-incomparable, then $(w_1, w_2) \in R_i \times S_i \subseteq \mathrm{Pref}$ asserts $w_2 \sqsubseteq w_1$, false. If they are comparable, say $w_1 \prec w_2$, then (w_1, w_2) asserts $w_2 \sqsubseteq w_1$, again false. □

Theorem 6.5 (B2: the intersection leaves Rat altogether)**.** *There are pushdown systems and states* β, α *for which* $\to^* \cap \mathrm{Pref}$ *is not a rational relation.*

Proof. The domain of a rational relation is a regular language. But $\pi_1(\to^* \cap \mathrm{Pref}) = L^{\beta\alpha}$, which is not regular by Theorem 3.2. □

So the failure is not a repairable defect in anybody's proof: no argument that stays inside the rational-transduction world can decide the prefix-restricted problem. The machine certificate is the Myhill–Nerode matrix for the instance of Theorem 3.2: for $C_0 = \{\alpha \times \beta \leq \alpha\}$ with the query $\beta \leq \alpha$, the words $0^i 1$ $(0 \leq i \leq 7)$ are pairwise inequivalent, separated by $0^j 1$, the test matrix being the identity (`topfree.py`).

Proposition 6.6 (B3: the theorem does not survive infinite rule sets)**.** *Theorem 6.1 requires* R *finite, and fails for rational* R*: in the mirrored (suffix) formulation of [8, Example 1, p. 6], the rational system* $R = (ab, a^2b^2) \cdot (b, b)^*$ *has* $\to^* (ab) = \{a^n b^n : n \geq 1\}$*, so* $\to^* \notin \mathrm{Rat}$.

Proposition 6.6 closes the other obvious repair. One would like to fold the prefix side condition into the rewriting system itself — let the system carry a component that remembers the initial stack and checks the prefix relation at the end. Any such system is infinite (rational at best), and then (1) is simply false. The obstruction is structural rather than technical: a pushdown system *consumes* its stack, whereas the prefix condition compares the final stack with the *initial* one, which the run has destroyed. That comparison is a property of the pair $(\text{start}, \text{end})$, not of the run — and Theorem 6.1 describes exactly that pair-relation, but in a class not closed under the comparison.

6.4 Opposite synchronizations, and why the residual is a periodicity

The three barriers have one common cause, which is worth isolating because it also explains the *shape* of everything downstream:

$\rightarrow^* = \mathcal{K}\cdot\Delta$ (common *suffix*: bottom of the stack), $\quad \mathrm{Pref} = \Delta\cdot(\Sigma^*\times\{\varepsilon\})$ (common *prefix*: top of the stac

Both are rational; each is "recognizable modulo one identity block"; but the identity blocks sit at *opposite ends* of the word. Imposing both on the same pair is precisely what produces a word equation: with $w = vzu$ and $v_f = vu$ one has $v_f \sqsubseteq w$ iff $u \sqsubseteq zu$ iff $u \sqsubseteq z^\omega$ (Lemma 4.1). In other words, Theorem 6.1 does not merely fail to decide the problem — it *predicts* that the residual object is a periodicity (cap) condition and not something wilder, which is the content of Theorem 4.2 and, downstream, the reason BorderAvoid is a question about borders and periods (Lemma 5.3) rather than about arbitrary comparisons.

Remark 6.7 (the same obstruction closes the MSO route)**.** Caucal's later work identifies prefix-recognizable graphs as those interpretable in the infinite binary tree, with decidable monadic second-order theory, and one would like to express BorderAvoid there. The obstruction recurs verbatim. A relation of the form $\mathcal{K}\cdot\Delta$ is MSO-definable in the tree whose ancestor relation is "is a suffix of": two vertices with a common ancestor and regular descents. But Pref is the ancestor relation of the *other* tree on the same vertex set, and interpreting one in the other is the copying that MSO cannot perform; the natural common expansion is the binary tree with the equal-level predicate, whose MSO theory is undecidable. So the MSO technology needs a new ingredient, not an assembly of existing ones.

Remark 6.8 (summary)**.** Theorem 6.1 yields, for this problem, exactly two things: the cap decomposition (Theorem 4.2) and the finiteness and shape of the flag alphabet (Proposition 6.2). Both are already exploited — the first is the residual object, the second is what makes §8 finite-state. By Proposition 6.4, Theorem 6.5 and Proposition 6.6 there is nothing further to extract: the open content of BorderAvoid, namely the bound on $\mathcal{B}$ of Theorem 8.9, lives strictly past the point where rational transductions stop being closed.

7 Structural results on BorderAvoid

These are statements about BorderAvoid, hence independent of the signature; proofs are as in the August 14 memo.

Proposition 7.1 (endpoint law)**.** *Every* BorderAvoid*-witness* π *(*$n = |\pi|$*) satisfies* $\neg G(c(i), c(n))$ *for every* $i \le n$*: the class of the final position must be* G*-incompatible with every class occurring anywhere in the word, itself included.*

Proof. Take $j = n$ in Lemma 5.1: the segment $\pi(i..n]$ has period $n-i$ vacuously (equivalently: every prefix of ρ carries the empty border). □

Lemma 7.2 (period-free extensions exist)**.** *For every $n, L \geq 1$ there is a word of length $\geq L$ over $\{0,1\}$ with no period $\leq n$: e.g. $(0^n1)^m$, whose least period is $n+1$.*

Lemma 7.3 (cleansing)**.** *Let $|\pi| = n$ and let x have no period $\leq n$. In πx no pair (i,j) with $j \leq n$ and $i < j$ satisfies the agreement condition of Lemma 5.2. Hence obligations anchored wholly in the past can always be discharged by appending a sufficiently aperiodic block; the obligations that cannot be outrun are those anchored in the final block.*

Proof. Agreement of (i,j), $i < j \leq n$, in πx makes $(\pi x)(i..]$ have period $p = j - i \leq n$; restricted to the last $|x|$ letters this gives x a period $p \leq n$. □

Proposition 7.4 (flag extremes)**.** *If G flags no class pair, every annotated word is a witness. If G flags all pairs, there is no witness: the pair (i,n) agrees for every i, violating Proposition 7.1.*

Proposition 7.5 (one-level cap checkability)**.** *Fix s, y, t with $y \neq \varepsilon$ and let $\pi_k = s\,y^k\,t$. It is decidable whether some π_k is a witness; indeed $\{k : \pi_k \text{ witness}\}$ is eventually periodic with effectively bounded threshold and period. The same holds for a fixed skeleton $s_0 y_1^{k_1} s_1 \cdots y_d^{k_d} s_d$ with the k_i free: all conditions become Presburger-definable in $(k_1, \ldots, k_d)$, using Fine–Wilf to reduce cross-block periodicity to linear congruences and threshold comparisons.*

What is *not* known is that witnesses can be normalized into a skeleton of bounded size. §8 replaces that question by a sharper one.

8 BorderAvoid in occurrence form

This section is the August 27 analysis, restated here as the current frontier. It is about BorderAvoid alone, so it applies verbatim to the $\top$-bearing signature as well.

Definition 8.1 (abstract BorderAvoid instance)**.** $\mathcal{I} = (A, K, \delta, q_0, \mathrm{Bad}, \mathrm{Fin}, G)$: a finite alphabet, a partial DFA, and $\mathrm{Bad}, \mathrm{Fin} \subseteq K$, $G \subseteq K \times K$. For $\pi \in A^*$ with $n = |\pi|$ and $c(i) = \delta(q_0, \pi[1..i])$, π is a *witness* iff no $c(i)$ is Bad, $c(n) \in \mathrm{Fin}$, and $s_j \sqsubseteq s_i \Rightarrow \neg G(c(i), c(j))$ for all $0 \leq i \leq j \leq n$.

By Theorem 2.6 and Corollary 2.8, Bad and Fin carry the conditions (S) and (F), and G carries the flag relation of §5.

8.1 Two coordinates instead of one

Proposition 8.2 (occurrence form)**.** *Let $U_\ell := \pi(n-\ell..n]$ be the length-ℓ suffix. For $0 \leq i \leq j \leq n$ put $\ell = n - j$ and $d = j - i$. The following are equivalent:* (a) *$s_j \sqsubseteq s_i$;* (b) *the length-$(\ell + d)$ suffix of π has period d;* (c) *U_ℓ occurs in π ending at position $n - d$;* (d) *$b := \ell$ is a border length of $\rho[1..m]$, $m := \ell + d$, where $\rho = \pi$ reversed and $c'(x) = c(n-x)$. Hence π is a witness iff*

(OCC) $\forall \ell, d \geq 0$ *with $U_{\ell+d}$ of period d :* $\neg G\big(c(n-\ell-d), c(n-\ell)\big)$, *i.e.* ***(BORD)*** $\forall m\, \forall b \in \mathrm{Bord}(\rho[1..m]$

Proof. (a)⇔(b) is Lemma 5.2; (b)⇔(c) reads a period as a self-overlap; (c)⇔(d) is Lemma 5.3. □

The point is that a constraint has *two* independent coordinates: the length $b = \ell$ of the repeated prefix of ρ and the period d it induces. Bounding one of them gives a finite-state condition; bounding either alone is insufficient (§8.4); bounding both is the sandwich.

Lemma 8.3 (prefix-closedness)**.** *(BORD) is prefix-closed: if ρ with annotation c' satisfies it, so does every prefix with the restricted annotation. The annotation is backward deterministic: $c'(m) = \delta(c'(m+1), \rho[m+1])$, $c'(n) = q_0$, $c'(0) = c(n)$.*

Corollary 8.4 (good tree)**.** *Let $T(\mathcal{I})$ be the tree of pairs $(\rho[1..m], c'(0..m))$ that are Bad-free, backward-consistent, start from $c'(0) \in$ Fin and satisfy* (BORD) *up to m. $T(\mathcal{I})$ is prefix-closed and finitely branching; witnesses are exactly its nodes with $c'(m) = q_0$; the shortest witness has length the least depth at which q_0 occurs. In particular a* finite *good tree is a complete NO certificate, detected in finite time.*

8.2 Fresh letters

Lemma 8.5 (fresh letter)**.** *If the last letter of π occurs exactly once in π, the only agreeing pairs are $i = j$ and $j = n$. Hence such a π is a witness iff $\neg G(c(i), c(i))$ and $\neg G(c(i), c(n))$ for all i — two conditions decidable by a subset construction on (current class, set of visited classes).*

Proof. Let $a = \pi[n]$ occur only at n, let $i < j < n$ and $s_j \sqsubseteq s_i$. Then s_j occurs starting at $i+1$, so its last letter a occurs at $i + (n - j) < n$: contradiction. □

8.3 The two-parameter sandwich

Definition 8.6. Fix $L, D \geq 0$. $N''(L, D)$ is (BORD) restricted to borders b of $\rho[1..m]$ with $b \leq L$ *or* $m - b \leq D$ (so $b = 0$ is covered by $b \leq L$ and $b = m$ by $m - b = 0 \leq D$), together with the Bad/Fin conditions. $S''(L, D)$ is $N''(L, D)$ together with

$$(*) \qquad \rho[1..L{+}1] \text{ has no occurrence in } \rho \text{ ending at a position } m > L{+}1{+}D.$$

Theorem 8.7 (sandwich)**.** *For all $L, D \geq 0$:*

(1) $N''(L, D)$ and $S''(L, D)$ define regular sets of annotated words, recognized by explicitly constructible automata with at most $|\text{Fin}| \cdot |A|^{2L+D+2} \cdot |K|^{L+D+3} \cdot 3^D \cdot (L{+}D{+}4) \cdot 2$ states; satisfiability of either is decidable.

(2) $S''(L, D)$ satisfiable $\Rightarrow$ $\mathcal{I}$ has a witness (and the automaton produces one).

(3) $\mathcal{I}$ has a witness $\Rightarrow$ $N''(L, D)$ is satisfiable, for every L, D.

(4) $\mathcal{I}$ has a witness $\pi \Rightarrow S''(L, 0)$ is satisfiable for $L = \text{lrs}(\pi)$, the length of the longest suffix of π that also occurs earlier; and $S''(0, D)$ for $D = n - e$, e the first occurrence-end of the last letter of π.

Proof. (1) Read ρ left to right, guessing the annotation backwards (Lemma 8.3). It suffices to keep: the first $L{+}1$ letters and $L{+}2$ classes (fixed after $L{+}1$ steps); the last $\max(L{+}1, D)$ letters and $D{+}1$ classes; for each $d \leq D$ one bit recording whether d is still a period of the current prefix; the position counter capped at $L{+}D{+}3$; one bit for $(*)$. The updates are local: d is a period of $\rho[1..m{+}1]$ iff it is one of $\rho[1..m]$ and $\rho[m{+}1{-}d] = \rho[m{+}1]$ (monotone), and

$b \le L$ is a border of $\rho[1..m{+}1]$ iff the last b letters equal the first b. The classes needed are $c'(b)$ for $b \le L$ (head window) and $c'(m-d)$ for $d \le D$ (tail window). (2) Assume $(*)$ and let b be a border of $\rho[1..m]$ with $b \ge L{+}1$ and $d = m - b \ge D{+}1$. Then $\rho[1..b]$ occurs ending at m, so $\rho[1..L{+}1]$ occurs ending at $m - b + L + 1 = d + L + 1 > L{+}1{+}D$, contradicting $(*)$. So every border has $b \le L$ or $m - b \le D$ and $N''(L, D)$ is the full condition (BORD); apply Proposition 8.2. (If $n \le L$, all borders have $b \le L$ already.) (3) $N''(L, D)$ is a sub-conjunction of (BORD). (4) For $L = \mathrm{lrs}(\pi)$, U_{L+1} occurs only once, i.e. $\rho[1..L{+}1]$ occurs only ending at $L{+}1$: that is $(*)$ with $D = 0$. For $L = 0$, $(*)$ with $D = n - e$ says the last letter of π does not occur before position e. □

Definition 8.8. $\mathcal{B}(\mathcal{I}) := \min\{L + D : S''(L, D) \text{ satisfiable}\}$ ($\min \emptyset = \infty$). By (2) and (4), $\mathcal{B}(\mathcal{I}) < \infty$ iff $\mathcal{I}$ has a witness, and $\mathcal{B}(\mathcal{I}) \le \mathrm{lrs}(\pi)$ for every witness π. $S''(0, 0)$ is Lemma 8.5.

Theorem 8.9 (what the open problem is)**.** BorderAvoid *is decidable if and only if there is a computable F with $\mathcal{B}(\mathcal{I}) \le F(|\mathcal{I}|)$ for every instance possessing a witness.*

Proof. ($\Leftarrow$) Test $S''(L, D)$ for all $L + D \le F(|\mathcal{I}|)$; answer YES on a success (sound by (2)), NO if all fail (a witness would force a success). ($\Rightarrow$) If BorderAvoid is decidable, search a witness π for each yes-instance and take $\mathrm{lrs}(\pi)$; the maximum over the finitely many instances of each size is computable. □

Remark 8.10. Theorem 8.9 replaces the two-conjecture route of the August 14 memo (witness normal form of bounded cap depth + bounded skeletons decidable) by a bound on a single integer, with a decision procedure that is a finite-automaton emptiness test: no normal form, no surgery, no Presburger translation. Since the bound is also necessary, nothing is lost by adopting it as the target. It also gives the programme its first finite refutation certificates: unsatisfiability of $N''(L, D)$, and finiteness of the good tree (Corollary 8.4).

8.4 The two extreme families, and why both are needed

Corollary 8.11 ($D = 1$: the final-run cap)**.** *Let π be a witness ending in the run a^r and put $x_s = \delta(c(n{-}r), a^s)$. Then $\neg G(x_s, x_t)$ for all $0 \le s \le t \le r$.*

Proof. For $n - r \le i \le j \le n$, $s_j = a^{n-j}$ is a prefix of $s_i = a^{n-i}$, so every such pair agrees, and $c(n{-}r{+}s) = x_s$. □

Since the a-orbit of a class is eventually periodic with index and period at most $|K|$, Corollary 8.11 caps r effectively: it is the $p = 1$ case of the capping phenomenon of `ba-capping-note`, with none of its bookkeeping.

Corollary 8.12 ($L = 1$: the last-letter law)**.** *Let π be a witness with last letter a. Then $\neg G(c(p), c(n{-}1))$ for every p with $\pi[p{+}1] = a$.*

Example 8.13. $A = \{0, 1\}$, $K = \{0, 1, 2\}$, $q_0 = 0$. *(i)* $\delta(q, 1) = \min(q{+}1, 2)$, $\delta(q, 0) = 0$, $\mathrm{Fin} = \{2\}$, $G = \{(0, 1)\}$. Every candidate ends in a run 1^r, $r \ge 2$, and the 1-orbit of 0 starts $0, 1, \ldots$ with $(0, 1) \in G$, so Corollary 8.11 refutes the instance at once — while all conditions with $b \le L$ are satisfiable for each fixed L (take $r > L{+}1$). *(ii)* $\delta(0, 0) = 2, \delta(0, 1) = 0, \delta(1, \cdot) = 2, \delta(2, 0) = 1, \delta(2, 1) = 0$, $\mathrm{Fin} = \{1\}$, $G = \{(0, 2)\}$. Reaching class 1 needs a final 0 from class 2, so $c(n{-}1) = 2$; by Corollary 8.12 no position of class 0 may be followed by 0, so from class 0 only 1 is available and $\delta(0, 1) = 0$: the run is trapped, and there is no witness — while the final run has length 1, so Corollary 8.11 says nothing.

Both examples were produced by an adversarial search explicitly hunting for instances the other family cannot settle.

8.5 The truncation theorem

Every surgery in the earlier programme (anchor cuts, in-root cuts, tower cuts, cycle-label replacement) excises material from the interior of a witness and must then argue that no new agreement is *created* at the seam; that creation analysis is where the remaining conjectures live. The following surgery creates nothing.

Definition 8.14. $e_1(\ell) := \min\{e \geq \ell : \pi(e-\ell..e] = U_\ell\}$, the first occurrence-end of the length-ℓ suffix. $e_1(\ell) < n$ exactly when $\ell \leq \mathrm{lrs}(\pi)$.

Theorem 8.15 (truncation)**.** *Let π be a witness, $0 \leq \ell \leq \mathrm{lrs}(\pi)$, $e := e_1(\ell)$, and suppose $c(e-\ell) = c(n-\ell)$. Then $\pi' := \pi[1..e]$ is a witness, $|\pi'| = e < n$, and $\mathrm{lrs}(\pi') < \ell$ for $\ell \geq 1$ (and $\pi' = \varepsilon$ for $\ell = 0$).*

Proof. The classes of π' agree with c on $[0, e]$. Since $\pi(e-\ell..e] = U_\ell = \pi(n-\ell..n]$ and the two occurrences start in the same class, $c(e-\lambda) = c(n-\lambda)$ for all $\lambda \leq \ell$; in particular $c(e) = c(n) \in \mathrm{Fin}$, and no class of π' is Bad.

lrs. By minimality of e, U_ℓ occurs in $\pi[1..e]$ only at the end. A suffix of π' of length $\lambda \geq \ell$ occurring earlier in π' would make U_ℓ — its own length-ℓ suffix — occur ending before e: contradiction.

Witness conditions. Let $\lambda \leq \mathrm{lrs}(\pi') < \ell$. The length-$\lambda$ suffix of π' is the length-λ suffix of U_ℓ, i.e. U_λ, so its occurrences in π' are occurrences of U_λ in π at the same positions; the required condition is $\neg G(c(\hat{e}-\lambda), c(e-\lambda)) = \neg G(c(\hat{e}-\lambda), c(n-\lambda))$, which holds because π is a witness. For $\lambda = 0$ this is $\neg G(c(p), c(e)) = \neg G(c(p), c(n))$, inherited from Proposition 7.1. For $\lambda > \mathrm{lrs}(\pi')$ the suffix occurs only once and only the diagonal pair remains, also inherited. □

Corollary 8.16 (rigid witnesses)**.** *Call π* rigid *if $c(e_1(\ell)-\ell) \neq c(n-\ell)$ for every $0 \leq \ell \leq \mathrm{lrs}(\pi)$. Every instance possessing a witness possesses a rigid one, and every witness of minimal length is rigid. Iterating Theorem 8.15 strictly decreases* lrs*; if it terminates at* $\mathrm{lrs} = 0$ *the result is a fresh-letter witness.*

Remark 8.17. Rigidity is a real constraint on a shortest witness: for every $\ell \leq \mathrm{lrs}(\pi)$ the pair $(c(e_1(\ell)-\ell), c(n-\ell))$ is simultaneously G-unflagged (because π is a witness) and off-diagonal (by rigidity); $\ell = 0$ gives $q_0 \neq c(n)$. Equivalently: the two runs of the class DFA entering the first and the last occurrence of U_ℓ never merge, for every ℓ up to lrs, while staying pointwise G-compatible. That is a statement about a pair of runs of *one* DFA on *one* word — no surgery, no creation analysis. A proof that rigidity cannot be maintained beyond $\ell = f(|K|)$ would, with Theorem 8.7(4) and Theorem 8.9, settle BorderAvoid. I did not find one.

8.6 Census

Verdicts of the procedure that tries all (L, D) with $L + D \leq B$ in increasing budget order (`occ_engine.py`); every S''-produced witness was re-verified against Definition 8.1, and no instance declared NO by N'' was ever found to have a witness by exhaustive search.

population	instances	max budget used	unsettle
random abstract, $\lvert K\rvert \le 6$, $B = 4$	4000	1	
random abstract, $\lvert K\rvert = 3, 4, 5$, $B = 4$ (period family only, $L = 0$)	24,000	3	2
adversarial hill-climb, $\lvert K\rvert = 3..7$, $B = 4$	12,500	2	
adversarial hill-climb, $\lvert K\rvert = 6, 8, 10$, $B = 5$	4,500	2	
SEP-derived, $\lvert K\rvert \le 29$, $B = 3$	200	1	
hard families A', $B'(m)_{m\le 6}$, C' of `rn-attempt`	18	1	

The second row is the one-parameter sandwich; its 22 unsettled instances are all settled at budget ≤ 2 by the two-parameter version, and 14 earlier "hard" instances — which resisted a depth-34 exhaustive search and the length-indexed family up to $L = 3$ — are all refuted by $N''(0, 1)$, i.e. by Corollary 8.11. The last row is the most informative: the families A', $B'(m)$, C' were built to defeat the *earlier* route, their minimal witnesses grow linearly ($5, 6, 9, 8, 11, 10, 13$ letters for $m = 0, \ldots, 6$) and $\lvert K\rvert$ grows from 9 to 15, yet $\mathcal{B} = 1$ throughout. So $\mathcal{B}$ is not driven by witness length — exactly the separation Theorem 8.9 needs. No instance surviving the two-parameter sandwich at budget 4 has been found.

Remark 8.18 (what the census does not show)**.** All of it lives at $\lvert K\rvert \le 29$, whereas the SEP-derived instances that matter have $\lvert K\rvert$ exponential in $\lvert C\rvert$ (the classes are subset constructions), and PSPACE-hardness makes short witnesses for *all* instances implausible. The census shows that the two families are individually incomplete and jointly strong, that $\mathcal{B}$ is insensitive to witness length on the programme's own hard families, and that every refutation found so far is by Corollary 8.11 or 8.12.

9 Conclusions

1. Over $\{\bot, \times\}$ a type *is* a prefix-closed set of paths and $\le$ *is* inclusion (Lemma 1.3); the order is a complete distributive lattice whose top is the type $\mu t.\, t \times t$ (Corollary 1.4). Removing the constructor $\top$ removes a label to be propagated, not a top element.

2. Consequently the path characterization needs one machine and one page (Theorem 2.6), instead of the two machines, safety analysis, permutation lemma and eight conditions (a)–(h) of `pda3.tex`. This completes `characterization.tex` (2003) and adds the two side conditions (S) and (F) that its "canonical substitution" sketch was missing. Countermodels are regular trees (Corollary 2.7). Verified: 300 random systems, four checks, 0 failures.

3. The residual object is unchanged. SEP over $\{\bot, \times\}$ reduces to a single instance of prefix-reach avoidance on $\mathcal{A}_\bot = \mathcal{W}^{\mathrm{rev}}$ with regular side conditions, hence to BorderAvoid (Theorems 2.10, 5.5). The regularity lemma that would have decided it is false, and the counterexample is realized $\top$-freely by $\{\alpha \times \beta \le \alpha\}$ with the query $\beta \le \alpha$ (Theorem 3.2).

4. BorderAvoid in occurrence form has two coordinates, and each gives a finite-state approximation from one side. BorderAvoid— hence SEP over $\{\bot, \times\}$ — is decidable iff the budget $\mathcal{B}$ is computably bounded (Theorem 8.9). The truncation theorem (Theorem 8.15) reduces that to bounding lrs for rigid witnesses, a statement about two runs of one DFA.

5. Caucal's theorem (§6) is what makes the flag alphabet finite, hence what makes the whole reformulation possible, and it is already fully exploited: it yields the cap decomposition and the finite flag shape and, by Propositions 6.4, 6.6 and Theorem 6.5, provably nothing beyond. Pref is not recognizable, $\rightarrow^* \cap$Pref is not even rational, and the theorem fails for infinite rule sets.

6. Not proved, and the whole content of the problem: a computable bound on $\mathcal{B}$. Nothing here shows it exists and nothing here shows it does not. Two concrete attacks are now available: bound lrs for rigid witnesses; or construct a family whose $\mathcal{B}$ grows faster than any elementary function of $|K|$ — testable mechanically, since `occ_engine.settle` computes $\mathcal{B}$ exactly.

A Reproducibility

All scripts are plain `python3`, no dependencies, in `analysis-2026/`.

- `topfree.py` — the machine $\mathcal{W}$, the conditions (S), (F), (B), the witness search, and the canonical countermodel of Theorem 2.6.
- `verify_topfree.py` — the four checks of §2.4 against exact enumeration of the antichain type family.
- `occ_engine.py` — the abstract BorderAvoid engine: exact witness test, the prefix-closed ρ-search of Corollary 8.4, the sandwich N''/S'' and $\mathcal{B}$, the truncation surgery.
- `occ_experiments.py` — the census and the verification of Lemma 8.5, Theorem 8.7 and Theorem 8.15 (26,743 truncations, 0 failures).
- `interestingattempt/caucal_shape.py` — the experiment of §6: intersection-shaped versus arbitrary flag relations; `interestingattempt/caucal-verdict.md` — the same verdict in note form, with the sources `caucal-1992.pdf` and `caucal-recritures-suffixes-de-mots.pd`
- `sep_bridge.py` — SEP instances (with $\top$) to abstract BorderAvoid instances, cross-validated against `ba_engine.py` (BEM pre*): 3840 tests, 0 mismatches.
- `ba_engine.py`, `ba_fast.py`, `verify_lemma3.py`, `verify_characterization.py` — from the August 14 session.
- `occ-form-note.pdf` — the stand-alone August 27 note on §8; `sep-status-and-attack.pdf` — the August 14 memo for the $\top$-bearing signature, of which this document is the $\top$-free revision.

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